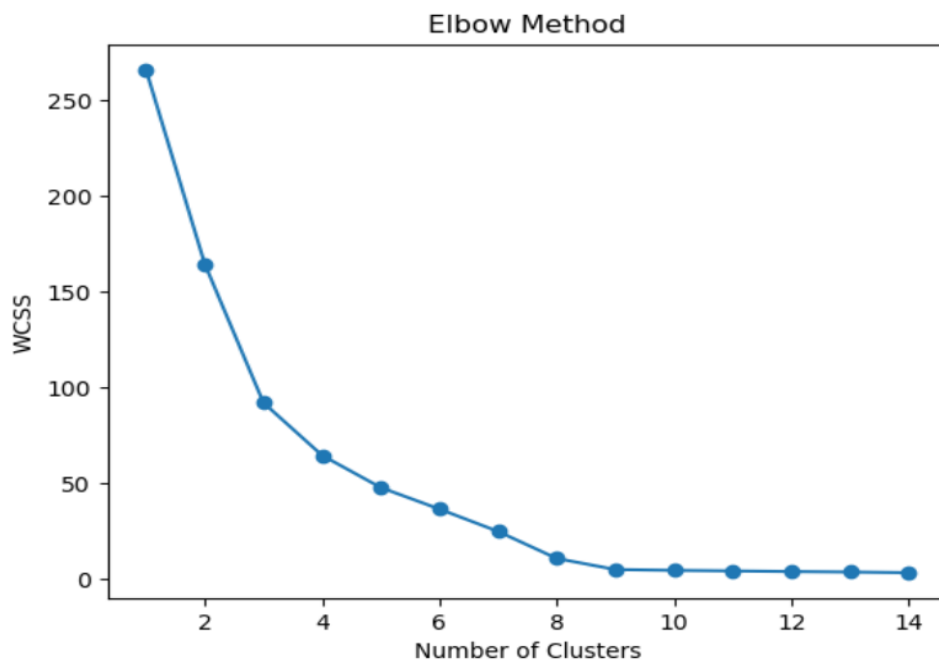
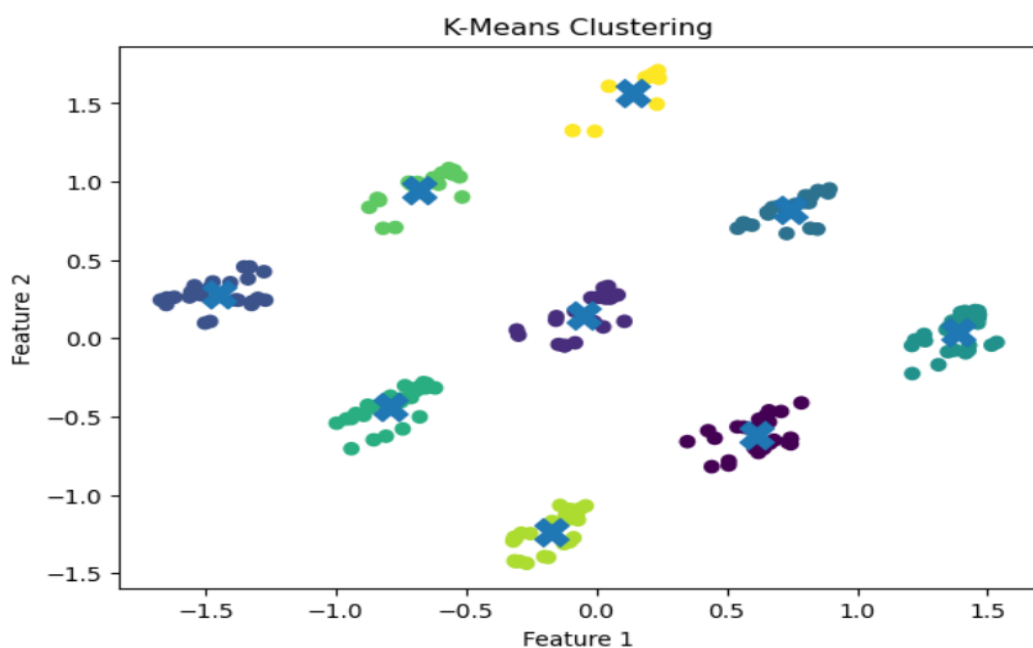


Elbow Method



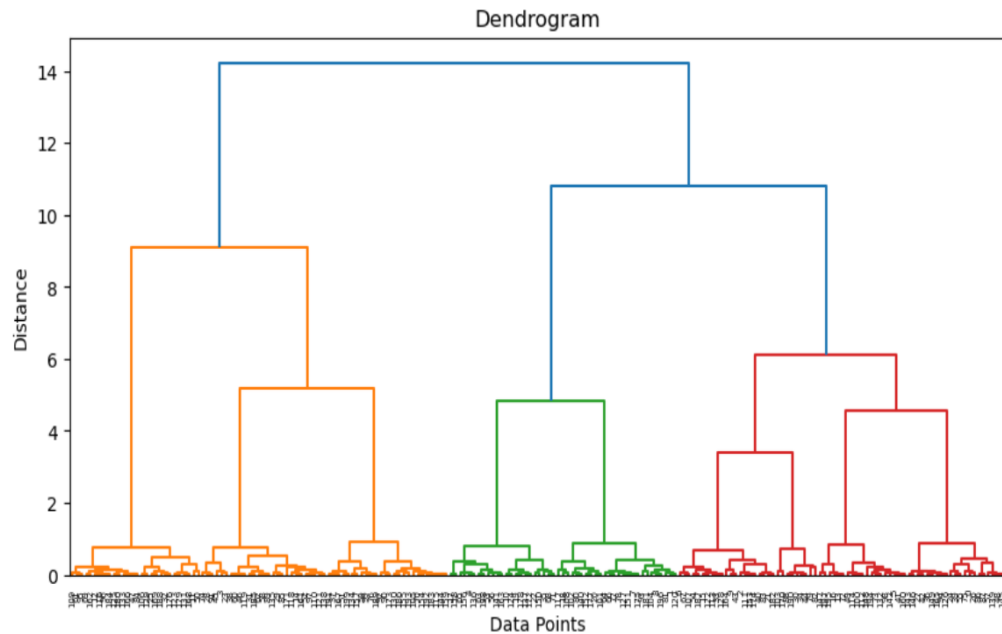
This plot helps determine the optimal number of clusters. The 'elbow' point (around $k=8$ or 9) indicates the best number of clusters where adding more clusters does not significantly reduce error.

K-Means Clustering



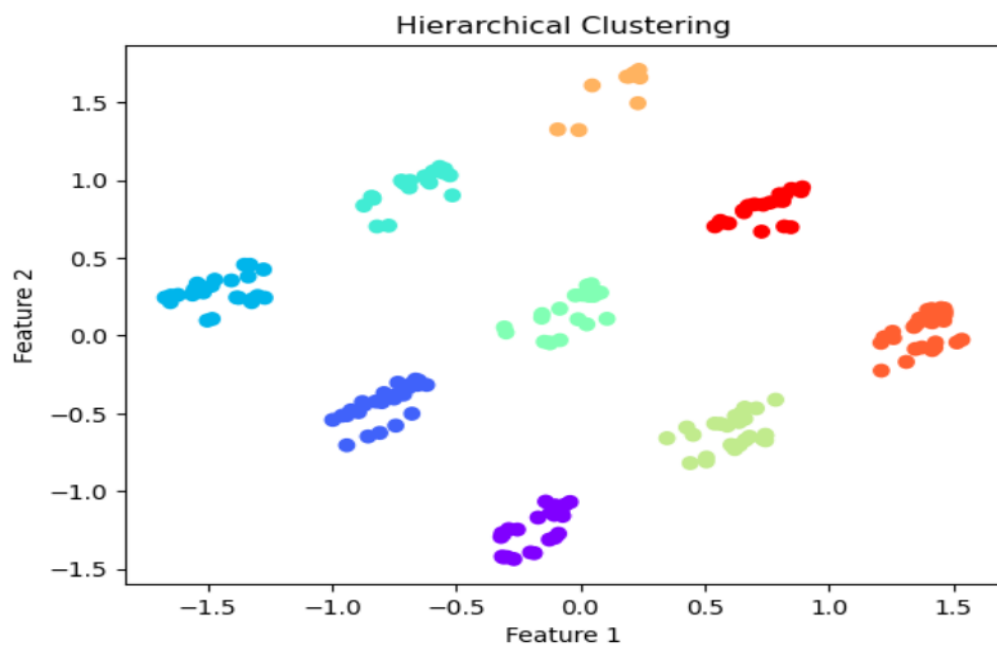
This plot shows how K-Means groups data into clusters. Each color represents a cluster, and the X marks represent centroids. The clusters are well-separated.

Dendrogram



This hierarchical clustering tree shows how data points are merged step by step. The vertical distance indicates similarity. Cutting the tree at a certain level gives the number of clusters.

Hierarchical Clustering



This plot shows clusters formed using hierarchical clustering. Similar to K-Means, the clusters are clearly separated, confirming strong grouping in the data.